FINANCE · CASE FORMULA

The financial case: $14M a year of problems, fixed for $9.4M

FIVE-YEAR NET IMPACT = INVESTMENT + 5 X (ADDITIONAL ANNUAL COSTS - ANNUAL SAVINGS)

$9.4M + 5 x ($1.3M - $6.4M) = -$16.1M net savings

WHAT THE MONEY FIXES (TARGETS FROM OUR PROBLEM STATEMENT)

YARD UTILIZATION
88% -> about 83% at +20% volume
target <= 85%

AVERAGE YARD INVENTORY
21,120 TEU -> about 19,850 TEU in year 5
target <= 20,400 TEU

CONTAINER DWELL TIME
6 days -> 4.7 days
target <= 4.83 days

TRUCK TURN TIME
72 min (40 queue) -> about 37 min
target < 45 min

INVESTMENT $9.4M OF $10.0M, BY ROOT CAUSE IT FIXES

METHODS	Yard zoning and block layout (MIP + CRAFT)	$1.2M	$100K/yr
METHODS	Dynamic slotting engine with demand forecasting	$1.6M	$300K/yr
TECHNOLOGY	Tracking and yard database (digital twin)	$2.2M	$350K/yr
EXTERNAL COORDINATION	Release visibility and truck pickup windows	$1.4M	$250K/yr
METHODS	AI AGV pilot: 2 units in a fenced lane	$1.6M	$150K/yr
PEOPLE	Change management and paid training	$600K	$150K/yr
RISK RESERVE	Contingency (about 9%)	$800K	$0/yr

Total $9.4M · $600K unspent **$1.3M/yr**

Initiative costs are our estimates (the case gives none); only the AGV vehicle price is anchored on a published figure. Phase 1 replaces them with fixed-price vendor quotes before the main spend.

ANNUAL SAVINGS $6.4M, BY ROOT CAUSE REMOVED

OVERTIME $2.4M of $6.0M (40%)

METHODS	Unnecessary rehandling, inefficient stacking	$1.20M
METHODS	Fixed placement rules: long travel	$600K
PEOPLE	Manual coordination, reactive re-planning	$600K

PENALTIES $4.0M of $8.0M (50%)

EXTERNAL COORDINATION	Poor information sharing: late release status	$2.04M
EXTERNAL COORDINATION	Unpredictable truck arrivals: 40-min queue	$1.28M
TECHNOLOGY	Limited tracking: boxes not ready on time	$680K
OUTSIDE OUR CONTROL	Customs holds and missing documents	not counted

Net $5.1M a year. Payback in about 22 months; break-even needs only $3.2M a year.

HOW EACH SAVING IS ESTIMATED

From root cause to dollars: the savings bridge

Saving = cost pool x share caused by the driver x share our lever removes. Shares and cuts are stated assumptions, checked against Phase 1 data before the main spend.

OVERTIME · $6.0M a year today — saved $2.4M (40%)

FISHBONE CAUSE	VSM STAGE	DRIVER	SHARE OF POOL	OUR LEVER	CUT	SAVING	OWNER
METHODS	Container retrieval	**Unnecessary rehandling, inefficient stacking**	40% = $2.4M	Dynamic slotting (stack by pickup order)	**50%**	**$1.20M**	Yard Planning Manager
METHODS	Yard allocation	**Fixed placement rules: long travel**	30% = $1.8M	MIP + CRAFT zoning (fast movers near exits)	**33%**	**$600K**	Yard Planning Manager
PEOPLE	Yard allocation	**Manual coordination, reactive re-planning**	30% = $1.8M	One yard database, one shared plan	**33%**	**$600K**	Operations Manager

PENALTIES · $8.0M a year today — saved $4.0M (50%)

FISHBONE CAUSE	VSM STAGE	DRIVER	SHARE OF POOL	OUR LEVER	CUT	SAVING	OWNER
EXTERNAL COORDINATION	Yard storage	**Poor information sharing: late release status**	30% = $2.4M	Release visibility on the database	**85%**	**$2.04M**	Commercial Director
EXTERNAL COORDINATION	Gate processing	**Unpredictable truck arrivals: 40-min queue**	20% = $1.6M	Truck pickup windows for ready boxes	**80%**	**$1.28M**	Gate & Landside Manager
TECHNOLOGY	Container retrieval	**Limited tracking: boxes not ready on time**	10% = $800K	Tracking + slotting (exact slot, staged boxes)	**85%**	**$680K**	Yard Planning Manager
OUTSIDE OUR CONTROL	Yard storage	**Customs holds and missing documents**	40% = $3.2M	Flagged early to customers; not counted	-	$0	Commercial Director

Total $6.4M a year = overtime $2.4M + penalties $4.0M. Customs holds (40% of penalties) are left out on purpose because PortFlow cannot override them.

UPFRONT COST AND RUNNING COST

Investment build-up: every dollar aimed at a root cause

INITIATIVE	ROOT CAUSE IT FIXES	VSM STAGE	UPFRONT	PER YEAR	BASIS OF THE NUMBER
Yard zoning and block layout (MIP + CRAFT)	METHODS Fixed placement rules	Yard allocation	$1.2M	$100K	Estimate
Dynamic slotting engine with demand forecasting	METHODS Inefficient stacking, rehandling	Container retrieval	$1.6M	$300K	Estimate
Tracking and yard database (digital twin)	TECHNOLOGY Limited tracking, fragmented systems	All stages (information gaps)	$2.2M	$350K	Estimate
Release visibility and truck pickup windows	EXTERNAL COORDINATION Poor information sharing, unpredictable trucks	Yard storage, gate processing	$1.4M	$250K	Estimate
AI AGV pilot: 2 units in a fenced lane	METHODS Manual horizontal transport	Ship unloading to yard	$1.6M	$150K	Vehicles sourced (~EUR 150k/unit, 2021); rest estimate
Change management and paid training	PEOPLE Manual coordination, reactive decisions	All stages	$600K	$150K	Estimate
Contingency (about 9%)	RISK RESERVE Cost overruns	-	$800K	$0	About 9% of the rest
TOTAL			**$9.4M**	**$1.3M**	**$600K under the $10.0M budget**

BUDGET ROOM FOR COST OVERRUNS

Without contingency the package is $8.6M. Costs can overrun by $1.4M (16%) before reaching $10.0M. Phase 1 gets fixed-price vendor quotes first.

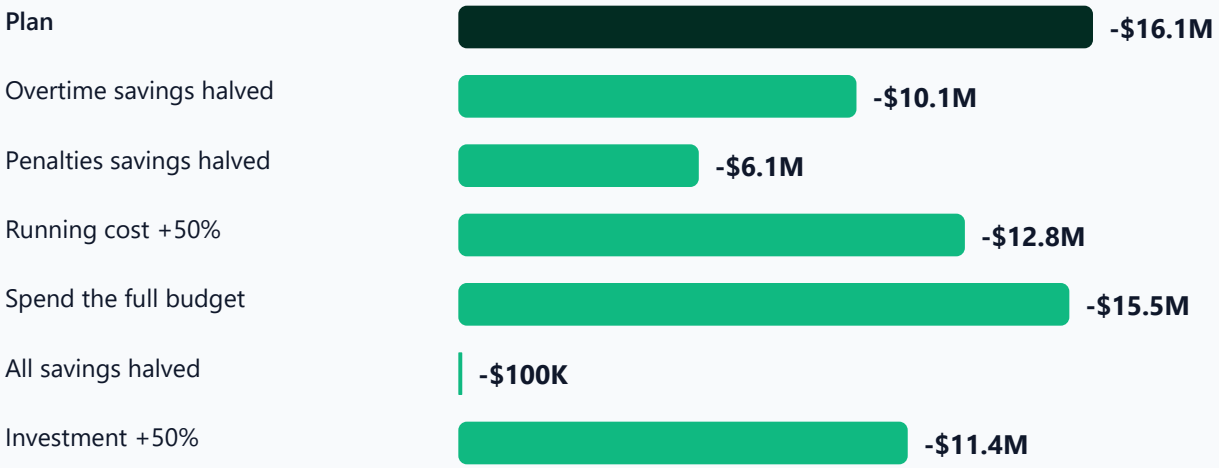
IF OUR COST ESTIMATES ARE WRONG

Even 50% higher investment ($14.1M) still gives -$11.4M over five years: savings drive the result far more than costs do.

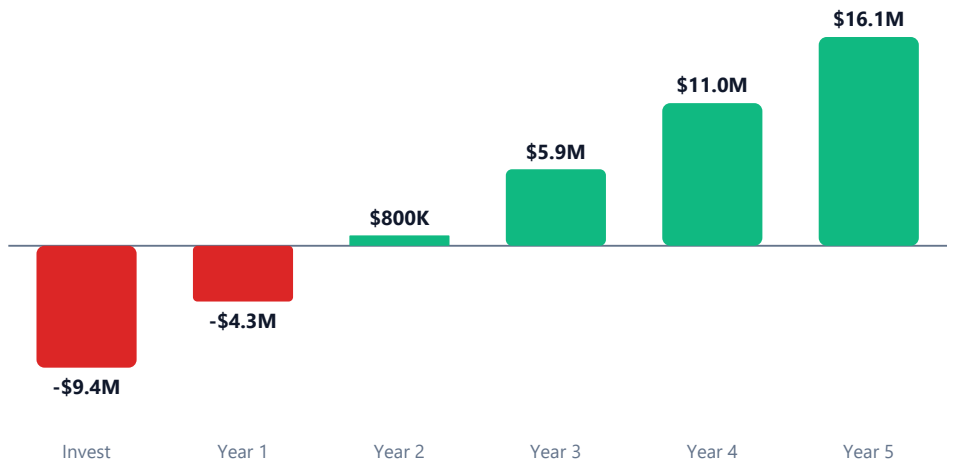
STRESS TEST

How robust is the result?

SAME FORMULA, ONE ASSUMPTION CHANGED AT A TIME



CUMULATIVE POSITION, UNDISCOUNTED



Every test stays a net saving. Break-even is $3.2M of savings a year (50% of plan). Payback on the case basis is about 22 months.

THE ONE FINANCIAL UNCERTAINTY

How much of the $8M penalty bill comes from customs or documentation holds that PortFlow cannot override. If only a quarter of penalties disappear instead of half, savings drop to $4.4M a year. Even then the result is -$6.1M. Phase 1 splits the penalty data by cause to confirm it.

GET READY TO PRESENT

Your 90 seconds, numbers and answers

SCRIPT WITH CUES (ABOUT 90 SECONDS)

[point at the formula bar]
Our problem costs $14M a year: $6M of overtime and $8M of penalties, driven by the root causes on our fishbone.

[point at the investment list]
We invest $9.4M of the $10M, and every dollar is aimed at one root cause: methods, technology, people or external coordination.

[point at the savings list]
That saves $6.4M a year: $2.4M of overtime from less rehandling, travel and re-planning, and $4.0M of penalties from better information, truck windows and ready boxes. Customs holds are not counted.

[point at the formula bar again]
Using the case formula: $9.4M plus 5 times ($1.3M minus $6.4M) equals minus $16.1M, a net saving.

[close on the robustness line]
It breaks even at $3.2M of savings a year, and even if costs ran 50% over we would still save $11.4M.

NUMBERS TO KNOW BY HEART

INVESTMENT **$9.4M of $10.0M**	RUNNING COST **$1.3M a year**
SAVINGS **$6.4M a year**	RESULT **-$16.1M (net savings)**
PAYBACK **about 22 months**	BREAK-EVEN **$3.2M a year**

LIKELY JUDGE QUESTIONS

Where do the costs come from?
Our estimates; the case gives none. Phase 1 gets fixed-price quotes, and even +50% costs still save $11.4M.

Why 40% and 50%?
Each comes from the savings bridge: cost pool x share caused by a fishbone driver x share our lever removes.

Why leave customs out?
PortFlow cannot override customs holds, so we do not claim those savings.

Why not discount?
The brief says to ignore discounting; the discounted check is still positive.

Did you count AGV labour savings?
No. It is a 2-unit pilot with no layoffs, so any AGV saving is upside.